

LESSON 6-2 • SESSION 1 • PRACTICE • FORM B

Dividing a Fraction by a Fraction

Learning target – I can divide a fraction by another fraction using the reciprocal. · 6.NS.A.1

REMEMBER

- To divide by a fraction, **multiply by its reciprocal**:
 $a/b \div c/d = a/b \times d/c$.
- Keep the first fraction, change $\div$ to $\times$, flip **only the second** fraction.
- Check**: quotient $\times$ divisor = the amount you started with.

WORKED EXAMPLE

A pitcher has $2/3$ liter of juice. Each cup holds $1/6$ liter. How many cups can be filled?

Step 1 Write the division: $2/3 \div 1/6$.

Step 2 Flip the divisor and multiply: $2/3 \times 6/1 = 12/3$.

Step 3 Simplify: $12/3 = 4$. Check: $4 \times 1/6 = 4/6 = 2/3$.

Answer 4 cups

WORDS reciprocal (*recíproco*) · dividend (*dividendo*) · divisor (*divisor*) · quotient (*cociente*)

PRACTICE

show your work · label units

1. Divide. Write each quotient in simplest form.

a. $7/8 \div 1/4 =$ _____

b. $3/4 \div 5/8 =$ _____

c. $10/11 \div 2/11 =$ _____

d. $4/5 \div 1/10 =$ _____

e. $1/4 \div 2/3 =$ _____

f. $5/9 \div 1/3 =$ _____

2. **Ink and Clay.** An art club has $5/6$ liter of ink and $6/7$ pound of clay. The bar is 1 liter; the shaded part is the ink.



a. Each bottle holds $1/18$ liter. How many bottles can the club fill? _____ bottles

b. Each clay portion is $2/7$ pound. How many portions? _____ portions

3. **Sanding Strips.** Theo has $9/10$ meter of sandpaper. He tears it into strips that are each $1/10$ meter wide. How many strips?

Hint Count the tenths in $9/10$.

Show your work.

Answer _____ strips

4. **Honey Jars.** A tub holds $3/4$ pound of honey. Each gift jar holds $1/16$ pound. How many jars can be filled?

Hint $1/4$ pound holds four $1/16$ pounds.

Show your work.

Answer _____ jars

5. **Wire Pieces.** A $3/8$ -meter wire is cut into pieces that are each $3/8$ meter long. Kofi says there are $9/64$ pieces. Justify or correct his answer.

Hint Did Kofi flip the divisor? How many $3/8$ s fit in $3/8$?

Show your work.

Answer _____ piece

6. **Wood Stain.** A can holds $1\frac{1}{12}$ liter of stain. Rosa uses $\frac{3}{12}$ liter on a shelf. She pours the rest into jars that each hold $\frac{1}{24}$ liter.

Hint Subtract first, then divide by $\frac{1}{24}$.

- a. How much stain is left? _____ L
b. How many jars does it fill? _____ jars

Show your work.

7. **Banner Ribbon.** A spool holds $\frac{4}{5}$ meter of ribbon. A large banner uses $\frac{2}{5}$ meter. A small banner uses $\frac{1}{10}$ meter.

- a. How many large banners can one spool supply? _____ banners
b. How many small banners? _____ banners
c. Explain why the second answer is 4 times the first. _____

8. **Name Tags.** A roll holds $\frac{2}{3}$ yard of trim. Each name tag uses $\frac{1}{15}$ yard of trim.

Hint A tiny divisor means many groups fit inside.

- a. How many name tags can the roll cover? _____ tags
b. Explain why the quotient is so large. _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Priya's Work.** Priya solved $\frac{4}{7} \div \frac{2}{3}$. She flipped both fractions and wrote $\frac{7}{4} \times \frac{3}{2} = \frac{21}{8}$. Explain her mistake. Then find the correct quotient.

TEST PRACTICE

choose the best answer

10. Find $\frac{2}{5} \div \frac{1}{10}$.

A $\frac{1}{25}$ B 4 C $\frac{1}{4}$ D 20

11. The reciprocal of $\frac{3}{7}$ is –

A $-\frac{3}{7}$ B $\frac{3}{7}$ C $\frac{7}{3}$ D $\frac{1}{7}$

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

2/3 E	8 S	9 I	2/45 X	3/64 Z	16 R	12 N	9/64 Q	1 V	10 E
#3		#4	#5	#6a	#6b	#7b	#8a		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____